

From open models to executable law

PolicyEngine and Axiom



Max Ghenis

Co-founder & CEO, PolicyEngine
Founder & CEO, Axiom Foundation

New technologies for evidence-based policy making
IARIW-CAPE workshop · UCLouvain Saint-Louis, Brussels · August 27, 2026



Tom just showed you the platform

BEAMM, EUROMOD, FPS Finance's models, PolicyEngine — the same craft, built by the same kind of people:

Encoded rules

Tax and benefit law, written as code that computes.

A representative population

Microdata that stands in for the country.

Validation

Checks against what the state publishes.

This talk is about where that stack is heading as AI joins the work — rules that carry their citations, populations calibrated to administrative totals, and verification that scales with the encoding. The code and the scoreboards are public.

The craft, scaled



I love building models. The work has always been:

- Program rules, encoded from the statute
- Data calibrated to what agencies publish
- Case studies that try to break the model
- Validation that catches when it does

WHAT CHANGES

Thousands of AI agents can now poke at every piece of that craft — encoding, testing, cross-checking — at a scale no individual analyst can match.



The question for the next 45 minutes: what does that scale make possible — and what does it not change?

What we'll cover



1 Rules

Law as executable code — the encoder, the Axiom Foundation, and a Belgian income tax validated against EUROMOD.

2 Data

Populations you can simulate on — imputation, calibration to administrative totals, and Microcosm-BE.

3 Live

Browse the encoded law, run a Belgian reform, score a policy in a few lines of Python.

4 Where this goes

Conductors, forecast uncertainty, and what it takes for these tools to enter public administrations.

1 Rules

Law as executable code — what exists today,
how AI encodes it, and how EUROMOD keeps
us honest in Belgium.

PolicyEngine today

Free, open source software to compute the impact of public policy. US and UK tax-benefit models, public code since June 2021 (AGPL-3.0), 133 contributors to the US model.

95,000+

parameters in the US model

5,500+

variables

4,693

test files

103

programs in the coverage registry

IN OTHER INSTITUTIONS' HANDS

No 10 Downing Street

The data science team built 10ds-microsim on PolicyEngine; CTO Nikhil Woodruff has served there as an Innovation Fellow since summer 2025.

NBER

MOU for an open source TAXSIM emulator.

Federal Reserve Bank of Atlanta

MOU bringing the Policy Rules Database into our validation infrastructure.

Think of it as an open EUROMOD-family stack for the US and UK, built in public.

Researchers, governments, and builders use these rules



AI cannot calculate taxes and benefits accurately today

POLICYBENCH.ORG · V1.1 · 2026-08-22

32 frontier models are scored on 100 real households — basic tax- and-transfer calculations, the building blocks of any microsimulation model.

1 in 9

amounts wrong to the dollar
best model — 88.7% exact (GPT-5.6 Sol)

>1 in 3

wrong
weakest model — 62.3% exact

On SNAP cases where the family is owed benefits, models mostly answer \$0 — none gets more than 1 in 20 right.

AI is impressive at code, search, and summarization. It is not a substitute for analyst judgment, institutional knowledge, or quality-control processes.

PolicyBench

Testing how accurately language models calculate household taxes and benefits.

32 MODELS / 100 HOUSEHOLDS / 18 OUTPUTS • SNAPSHOT 2026-08-22

LEADERBOARD

Model rankings

SERVING SENSITIVITY · AUGUST 2026

Claude models skip extended thinking when the answer tool call is forced, as it is in the identical request this board holds every model to; other reasoning-by-default providers reason regardless. Re-run with `tool_choice: auto`, Claude Fable 5 scores 86.9 (would rank #2), Claude Opus 5 85.6 (#3), and Claude Sonnet 5 80.2 (#8). The board below is unchanged — those runs sit beside it as a labeled sensitivity — and the next board version moves every model to auto.

#	MODEL	RELEASED	COST / HOUSEHOLD	EXACT
1	 GPT-5.6 Sol	2026-07-09	\$0.090	88.7%
2	 Claude Fable 5	2026-07-09	\$0.075	86.9%
3	 Claude Opus 5	2026-07-09	\$0.075	85.6%
4	 Claude Sonnet 5	2026-07-09	\$0.030	80.2%

But it can help us encode them faster



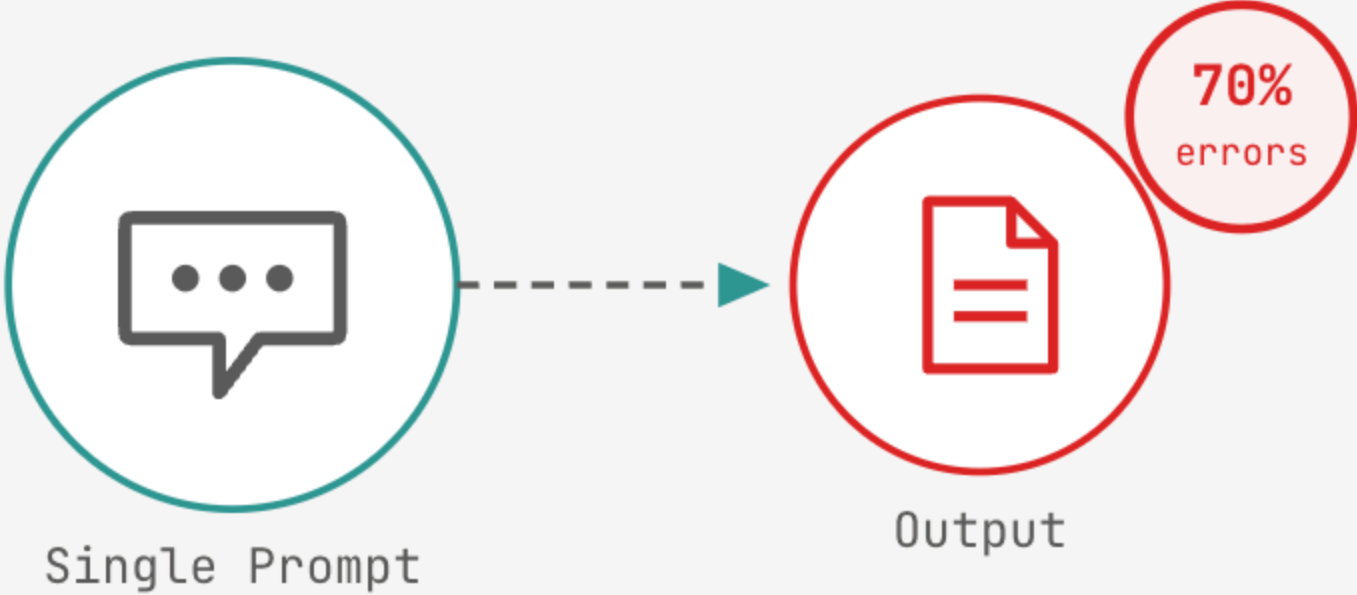
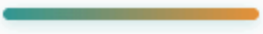
AI can't accurately compute policy on its own. **It can help us build the substrate that does.**

Over the last three years we have evolved how AI assists with the encoding work itself — from a single prompt that often failed to a multi-agent pipeline that produces tested, citation-backed PRs.

Six stages of evolution, six lessons learned, one workflow we use today.

Evolution step 1: single prompt

70% error rate



We started with the simplest approach: a single prompt asking Claude to implement one benefit program end-to-end. **One command, one output**—parameters, variables, tests, everything in one shot.

What worked

- ✓ Simple to start—one prompt
- ✓ Produces accurate code on simple tasks
- ✓ Fast iteration cycles

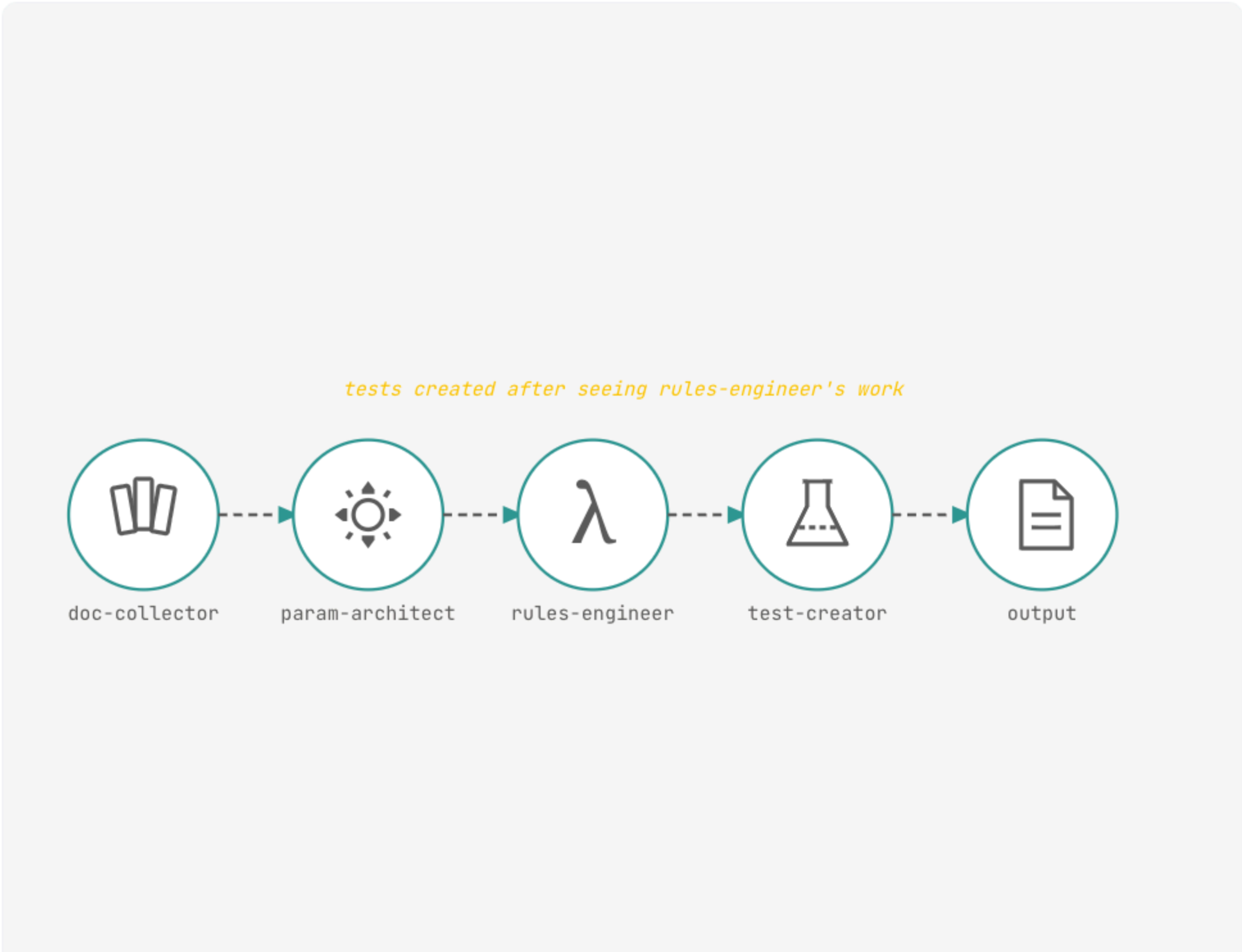
What didn't

- ✗ 70% of outputs had errors
- ✗ Hard-coded values
- ✗ No source citations

Key insight: Too much context for one prompt. The model couldn't hold code standards, legal requirements, and testing patterns all at once.

Evolution step 2: core pipeline

4 agents, sequential



We introduced **agents**—specialized AI workers that each focus on a single concern. A document-collector gathers sources. A parameter-architect structures data. A rules-engineer implements logic. A test-creator writes validations.

What worked

- ✓ Each agent focused on one task
- ✓ Clearer, shorter prompts
- ✓ Easier to debug failures
- ✓ Improve agents independently

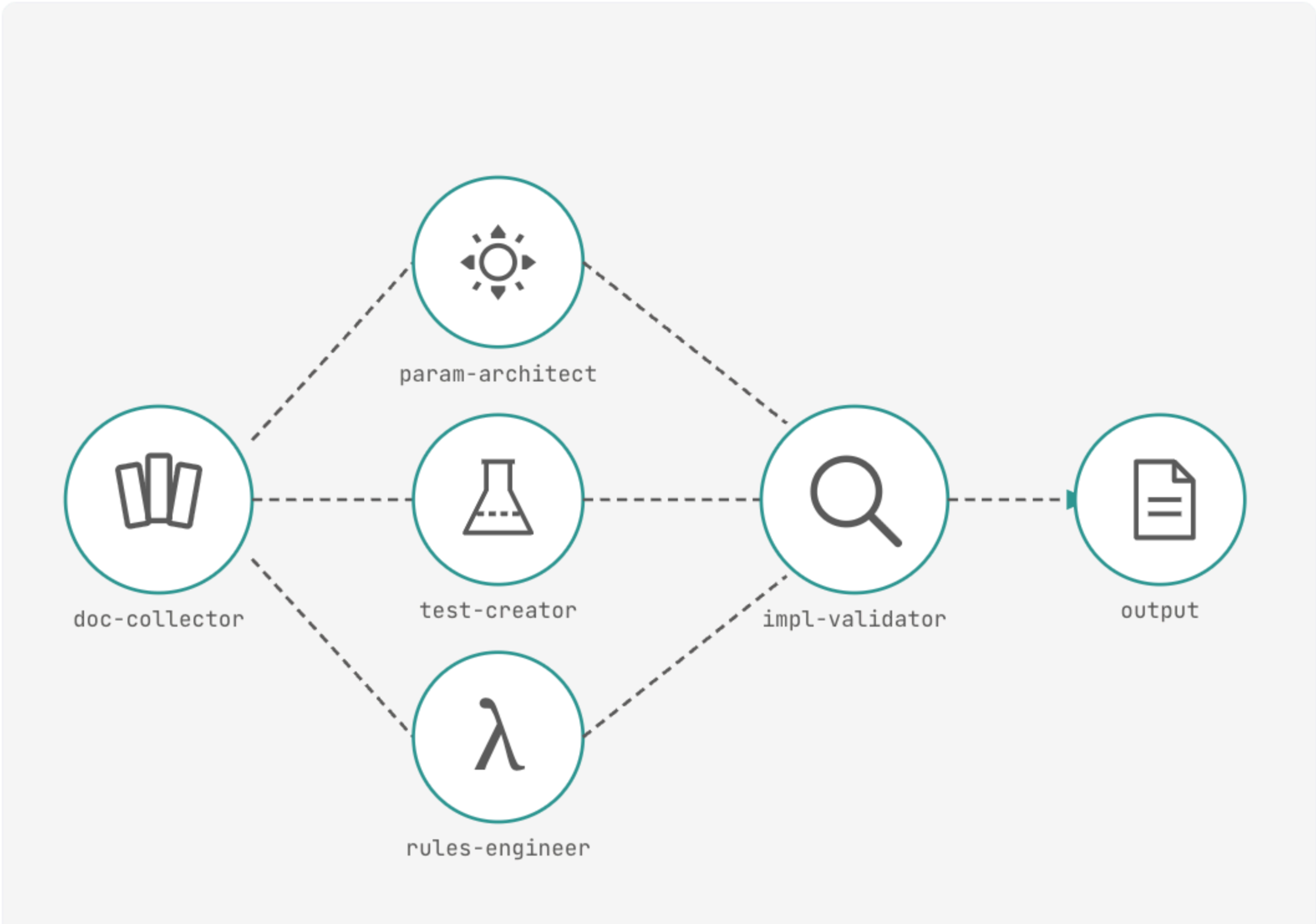
What didn't

- ✗ Tests run with variable logic in mind
- ✗ Bugs are reinforced, not fixed
- ✗ No validation step

Key insight: Sequential execution means tests validate existing code patterns—not actual program document requirements.

Evolution step 3: parallel execution

5 agents, 40% error rate



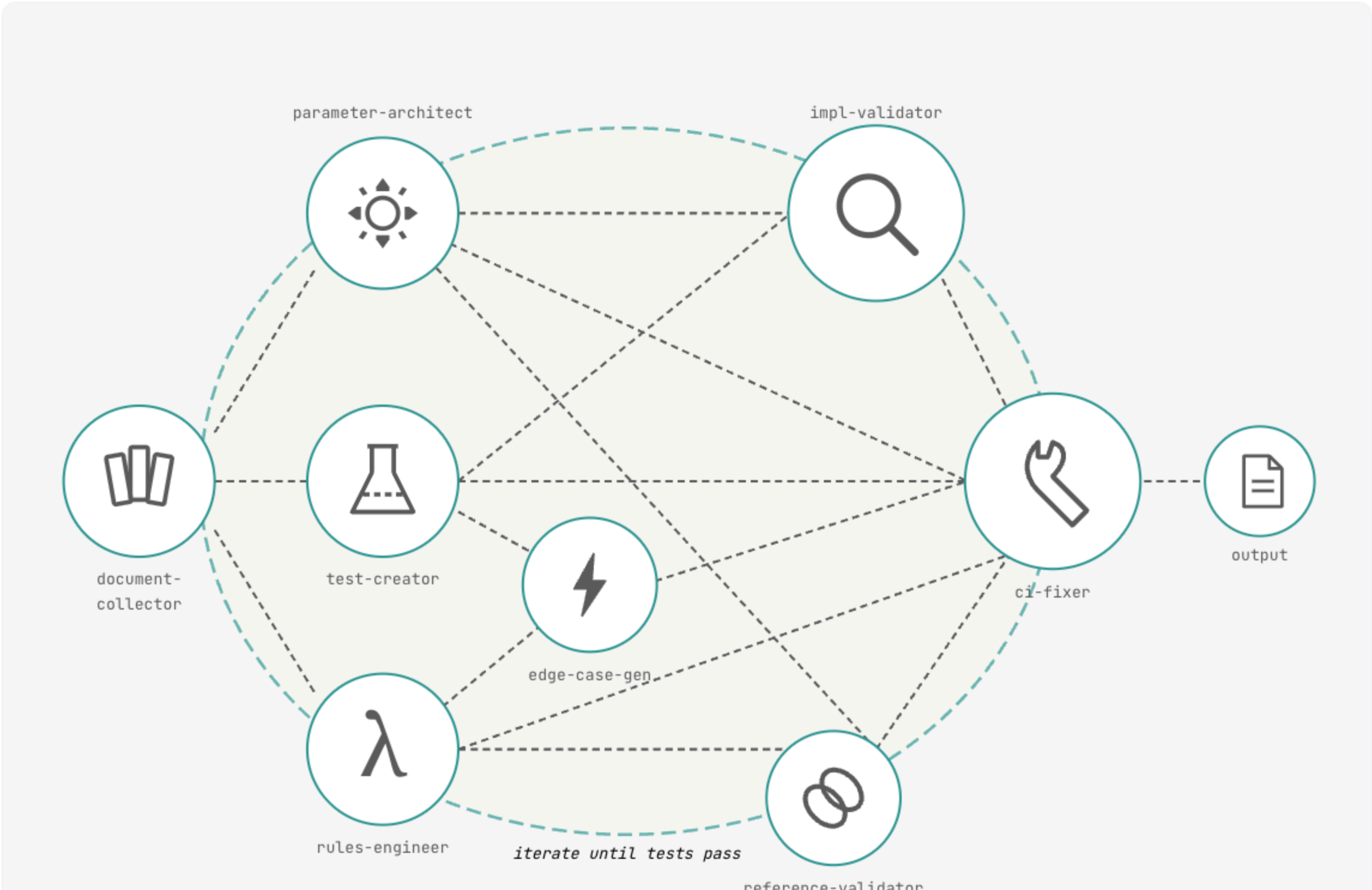
Test-creator and rules-engineer run in **parallel**. Both read the same documentation. Neither sees the other's output. When tests fail, it reveals actual discrepancies between implementation and requirements.

What worked	What didn't
✓ Tests verify requirements	✗ No reference validation
✓ Catches real bugs	✗ No edge case testing
✓ Error rate dropped to 40%	✗ Manual review iterations
✓ Added impl-validator	✗ Styling mismatch between params and vars

Key insight: Agent isolation prevents confirmation bias but requires manual intervention on each iteration.

Evolution step 4: validation loop

8 agents with CI feedback



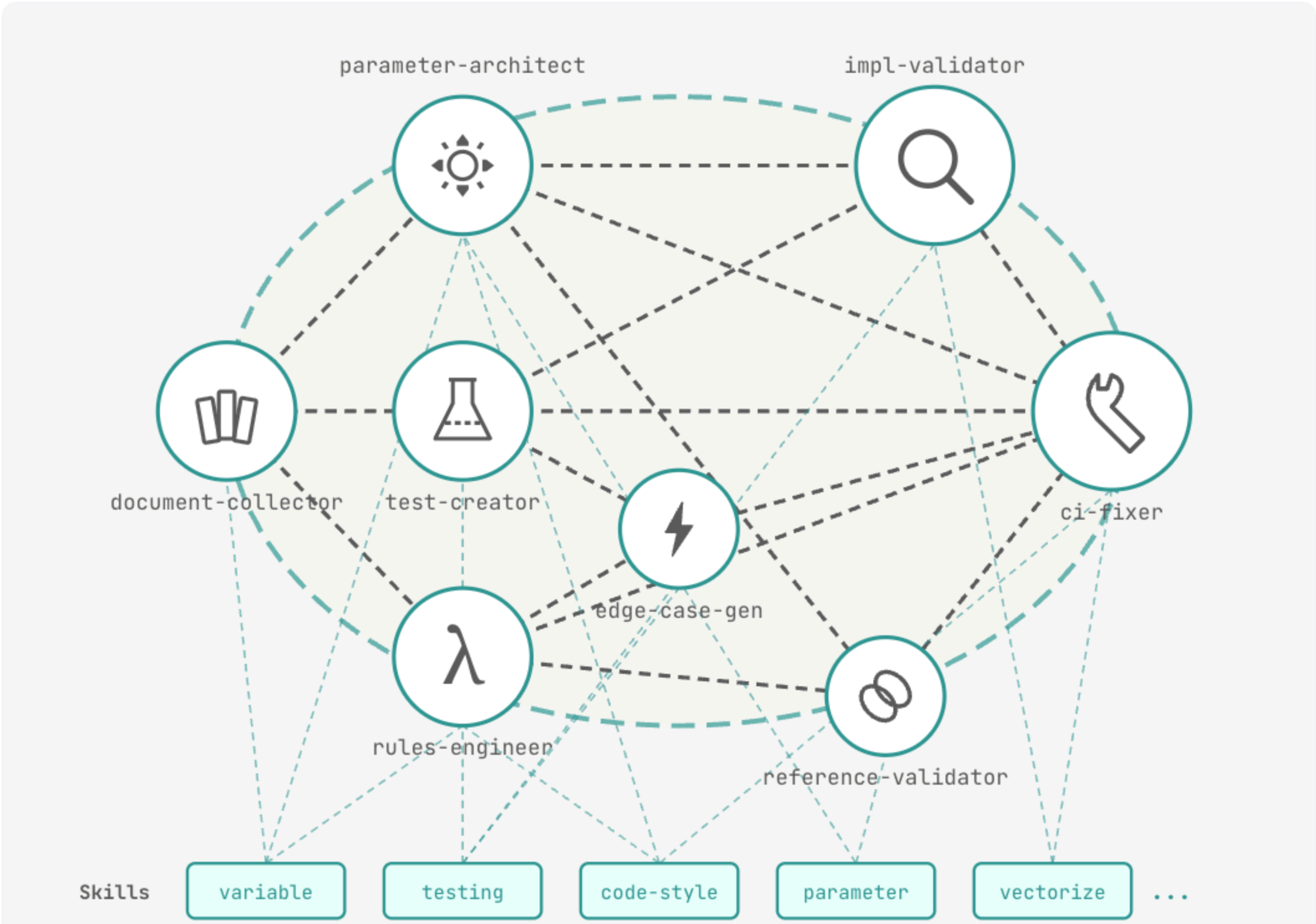
We added **edge-case-generator**, **reference-validator**, and **ci-fixer** to automatically iterate on failures. The workflow now loops—when tests fail, ci-fixer analyzes the error and retries.

What worked	What didn't
✓ Automatic iteration cycles	✗ Inconsistent patterns
✓ Edge cases covered	✗ Long repeated prompts
✓ Citations validated	✗ Same rules in every agent
✓ Less manual work	

Key insight: More agents provide more coverage, but also introduces more inconsistency. Shared knowledge is necessary at scale.

Evolution step 5: modular skills

15% error rate — reusable knowledge modules



We introduced **skills**—reusable knowledge modules that agents load on demand. Instead of repeating instructions in every prompt, we extracted shared knowledge into focused documents.

variable-patterns
Proper class structure

testing-patterns
Consistent test format

code-style
Formatting rules

parameter-patterns
YAML file structure

vectorization
NumPy operations

review-patterns
Review checklist

70%

Step 1 errors

40%

Step 3 errors

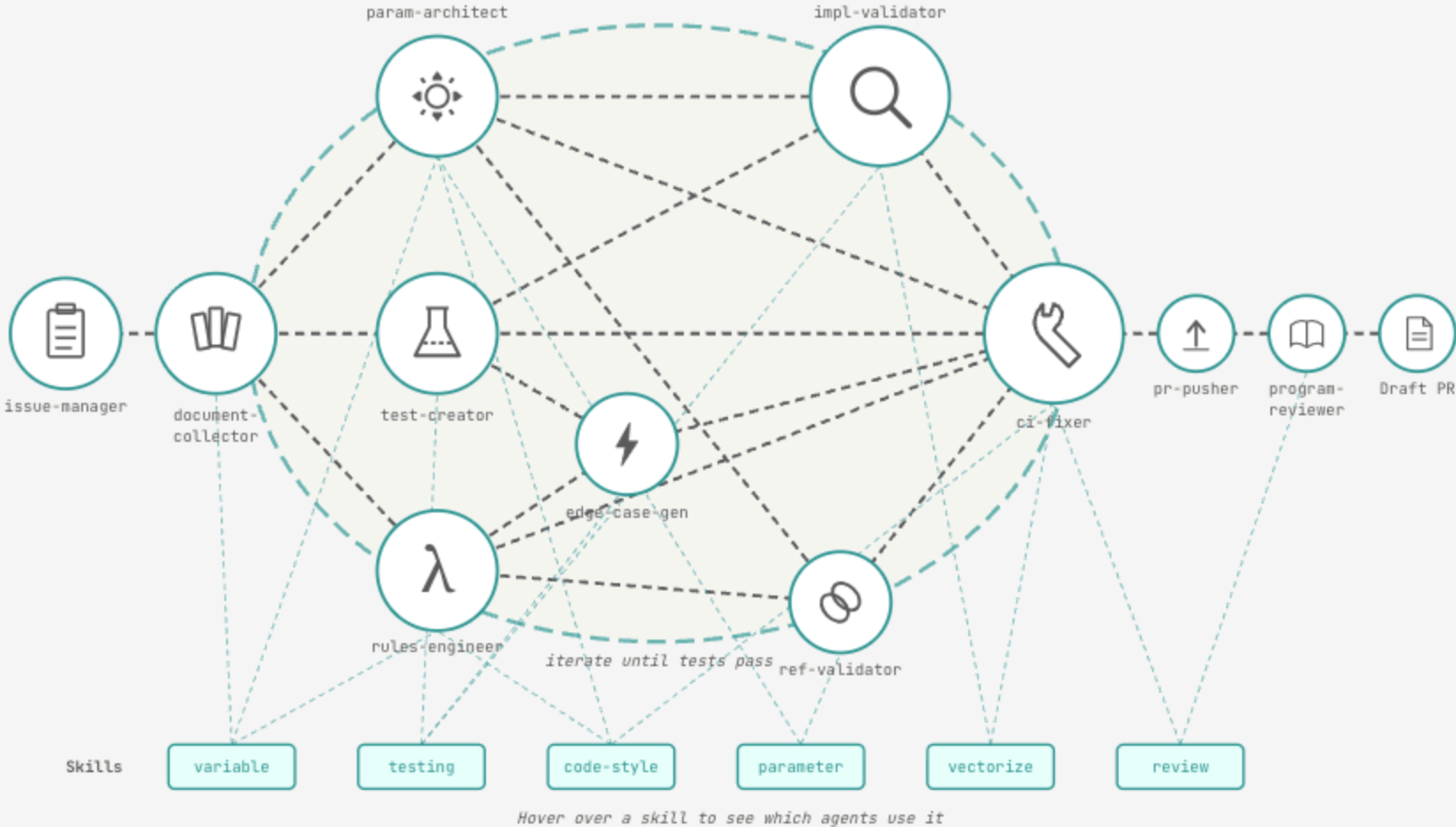
15%

Step 5 errors

Key insight: Skills let us scale without redundancy. One skill, many agents—consistent patterns everywhere.

Evolution step 6: full workflow

Complete pipeline — one command to a draft PR in ~90 minutes



The complete pipeline: **12 agents** and **6 skills** working together. One command triggers the entire workflow—from issue creation to a draft PR ready for human review.

What it produces: 11 parameter files, 8 variable files, 65 test cases, 1 complete PR—all in ~90 minutes.

Human role: Review the final PR and make the merge decision. Agents handle all mechanical work.

~90

minutes

12

agents

6

skills

The orchestrator never writes code—it invokes agents and checks quality gates.

The Axiom Foundation: encode, verify, publish

1

Accuracy at every step

Every parameter, formula, citation, and test produced along the way — not only the final answer.

2

Cross-validation against other sources

TAXSIM, agency tables, peer modelers — multiple ground truths beyond our own.

3

Deterministic quality gates

Tests, lint, type checks, structural rules that the agents must pass — not opinions about quality.

4

Human review at the right boundary

Not every prompt. The final PR, the assumptions, the edge cases worth a second pair of eyes.

5

A canvas wide enough for the work

A queue that can hold thousands of policies and run them in parallel without losing context.

THE WORLD'S RULES, ENCODED

Encode

Statute to RuleSpec. Every value cites its authority; every clause carries its effective dates.

Verify

A deterministic gauntlet, then oracles that cross-check against external engines and datasets — PolicyEngine, TAXSIM, EUROMOD, SNAP quality-control data — so the model that wrote the rules never grades its own work.

Publish

Source document, encoding, validation record, and computation graph in one place.

axiom.org

Belgian income tax, validated against EUROMOD

23/23

in-scope BE_2025 policies conformant

156

household comparisons across 33 suites

0

unexplained mismatches

EVERY DISAGREEMENT DISPOSITIONED

The suites raised 40 raw disagreements. 33 traced to upstream engine differences and are filed as findings for the EUROMOD team's review on the public EC-JRC tracker; 7 are explained residuals.

EUROMOD is the reference we are glad exists — the cross-checks flow both directions.

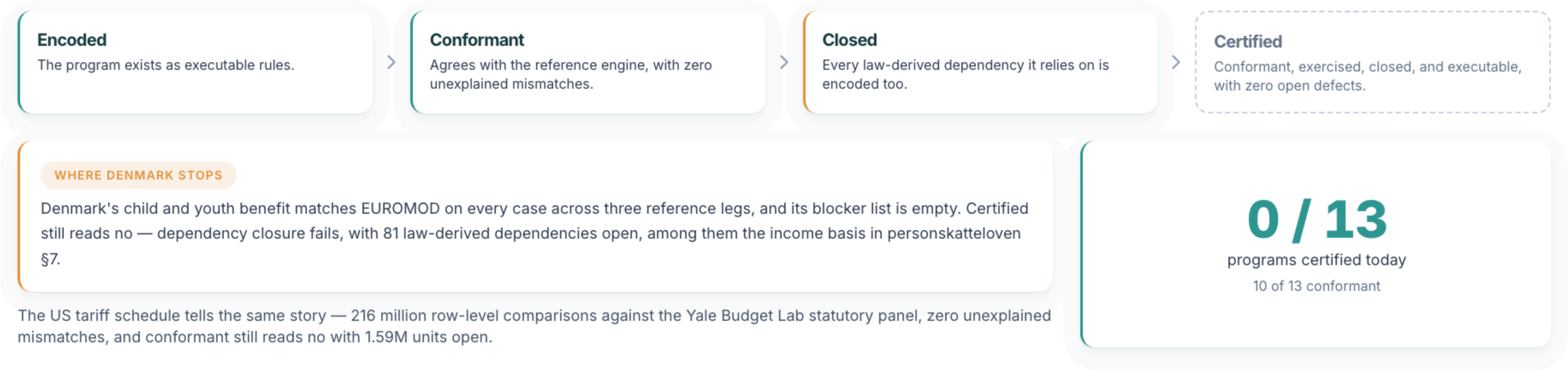
WORKED EXAMPLE

The Belgian PIT arc — couples, pensions, self-employment, and replacement-income pipelines — is merged on main. Dependants and child benefits are encoded, held at the signed-corpus release frontier.

rulespec-be and the oracle harness are public at github.com/TheAxiomFoundation · scoreboard at axiom.org/oracles

Complete is not certified

Agreement with the reference engine is necessary — and nowhere near sufficient.



Certified is computed, never set by hand, and the registry publishes whatever the computation says. The scoreboard grades itself honestly — that is the point.

2 Data

Populations calibrated to administrative truth —
and what that looks like for Belgium.

Building a dataset that is comprehensive and accurate

To represent every household's tax and benefit position, the microdata needs to be **comprehensive** enough to cover the relevant variables and **accurate** enough to reproduce administrative totals.

THE US CONSTRAINT

No single public survey has everything. The CPS misses capital gains. The SCF misses program participation. Administrative data is locked behind privacy walls we cannot link through.

THE EUROPEAN INVERSION

In Belgium the registers are already linked. The binding constraint is release, not linkage — which is why BEAMM works on synthesis, building populations the public can use.

Whether you start from survey fragments or from locked registers, both directions converge on the same need — calibration targets and validation services.

AI has accelerated both halves of this work — from understanding what each source actually contains, to ingesting the growing universe of administrative targets, to rethinking the whole pipeline from scratch.

Imputation at scale: many sources, one microdata

SOURCES WE FUSE IN THE US MODEL

CPS **Current Population Survey**
Spine — demographics, income, labor force.

ACS **American Community Survey**
Geography, housing, sub-state detail.

SCF **Survey of Consumer Finances**
Wealth, capital income, debt.

SIPP **Survey of Income and Program Participation**
Program take-up, dynamics, transitions.

PUF **IRS Public Use File**
Tax-unit income detail, itemized deductions.

TECHNIQUE

Quantile regression forests. For each variable missing from the spine, learn its full conditional distribution given many potentially non-linear predictors — then sample from it.

The whole distribution survives, including the tails that matter for distributional analysis.

WHERE AI HELPS

The hard part is knowing what each column actually means — the same name across two surveys often hides different definitions. AI reads the documentation, surfaces mismatches, and proposes the apples-to-apples mapping. Get this wrong and the imputation silently drifts.

Calibration: matching administrative totals

THE METHOD

Gradient descent over household weights, optimizing simultaneously against published administrative targets — tax authority tables, statistical office series, census tabulations, program totals.

WHY IT WORKS

When the optimizer fits all targets at once, the resulting weights reproduce the cross-tabulations ministries and statistical offices already publish — while preserving the joint distribution the microdata carries.

WHERE AI CHANGES THIS LAYER

- **Target ingestion.** Agents read statistical releases, tax authority tables, and ministry PDFs and pull numbers into a canonical target format, keeping pace as publications land.
- **Schema reconciliation.** When definitions change between releases, agents surface the diff before it silently miscalibrates the model.
- **Promotion decisions.** The agent proposes which targets to add; analysts decide which ones go into the live calibration and which stay in reserve.

Same calibration discipline. Many more targets, much faster turn, humans on the judgment calls.

Microcosm-BE: a calibrated Belgian population

57,240

household records

254

calibration targets

1.84%

mean absolute error

242/254

targets within 5%

v0.5 release. Targets are sums of Chronicle facts — Belgian administrative and national-accounts data. Surveys enter as validation only.

Every EUROMOD output column accounted for

All 182 substantive EUROMOD output columns sit in a ledger on this release: 10 matched to 1e-9, 143 explained, 29 named gaps, 0 unclassified.

Same population, two engines

Joint-vs-separate PIT on the identical calibrated population (v0.2): €2.16B in Axiom, €1.83B in EUROMOD. The residual is decomposed mechanism by mechanism.

PROVENANCE

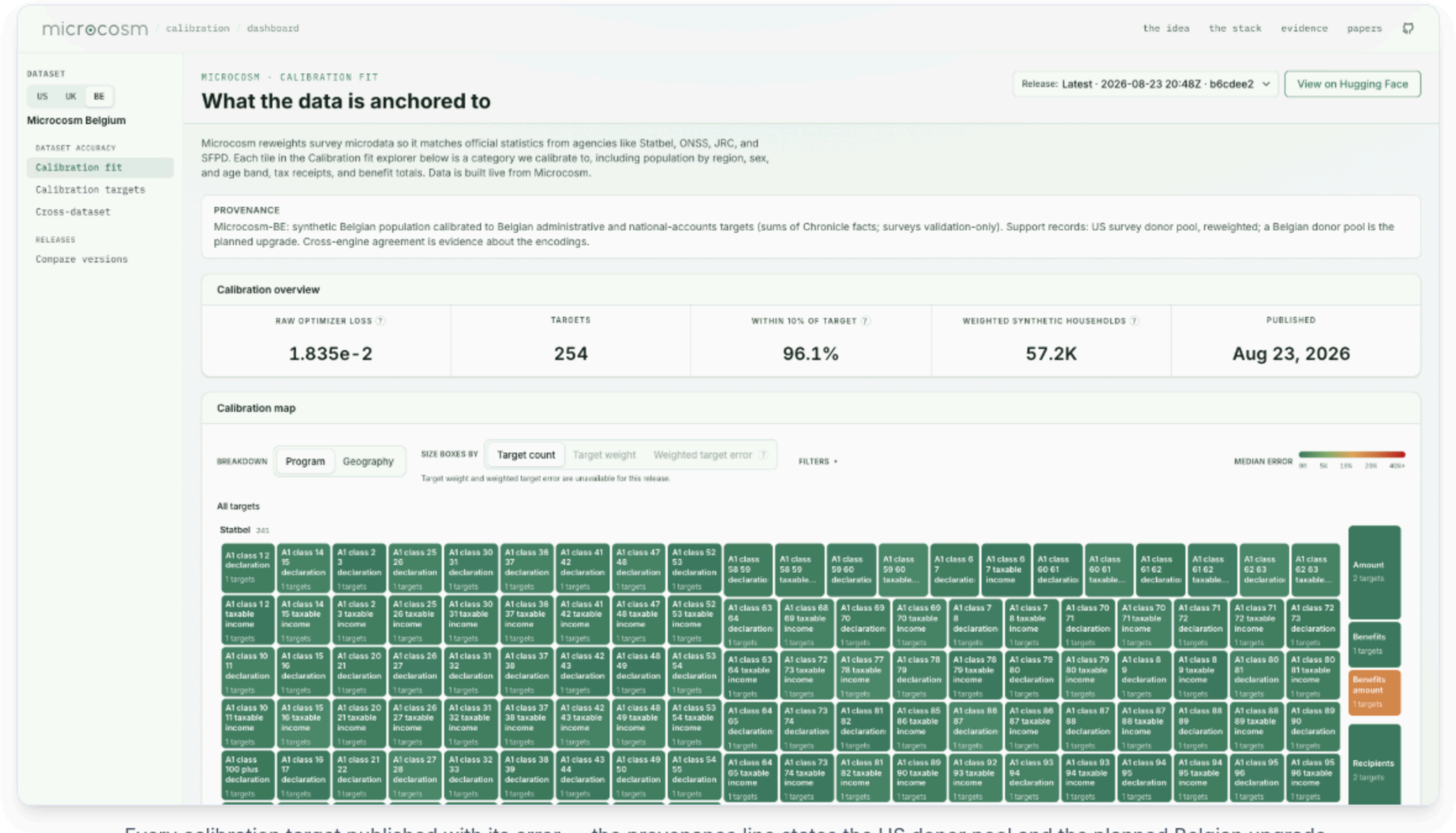
Support records are a reweighted US survey donor pool; a Belgian donor pool is the planned upgrade.

LIVE

The calibration dashboard at microcosm.institute serves this release publicly.

The data, anchored in public

microcosm.institute/calibration/dashboard



Every calibration target published with its error — the provenance line states the US donor pool and the planned Belgian upgrade.

Microplex: rethinking the whole pipeline

We are replacing Enhanced CPS, our US microdata product, with a new paradigm we call **Microplex**.

Same imputation + calibration discipline. Different architecture — designed so the next decade of survey changes, administrative releases, and program reforms can flow through without rebuilding from scratch.

AI is what makes this scope tractable. Without it, rebuilding the substrate at this depth would take many years of analyst time. With it, we can iterate releases on a quarterly cadence.

Multi-spine, not single-spine

CPS and ACS combined as the foundation, with structured ingestion from each — rather than treating CPS as the spine and bolting everything onto it.

Automated target discovery

Agents continuously crawl agency publications for new administrative totals, structured into the canonical target format.

Promote-on-review

New targets land in a reserve queue. An analyst decides whether each becomes a live calibration target or stays as reference information.

Replicable, versioned, public

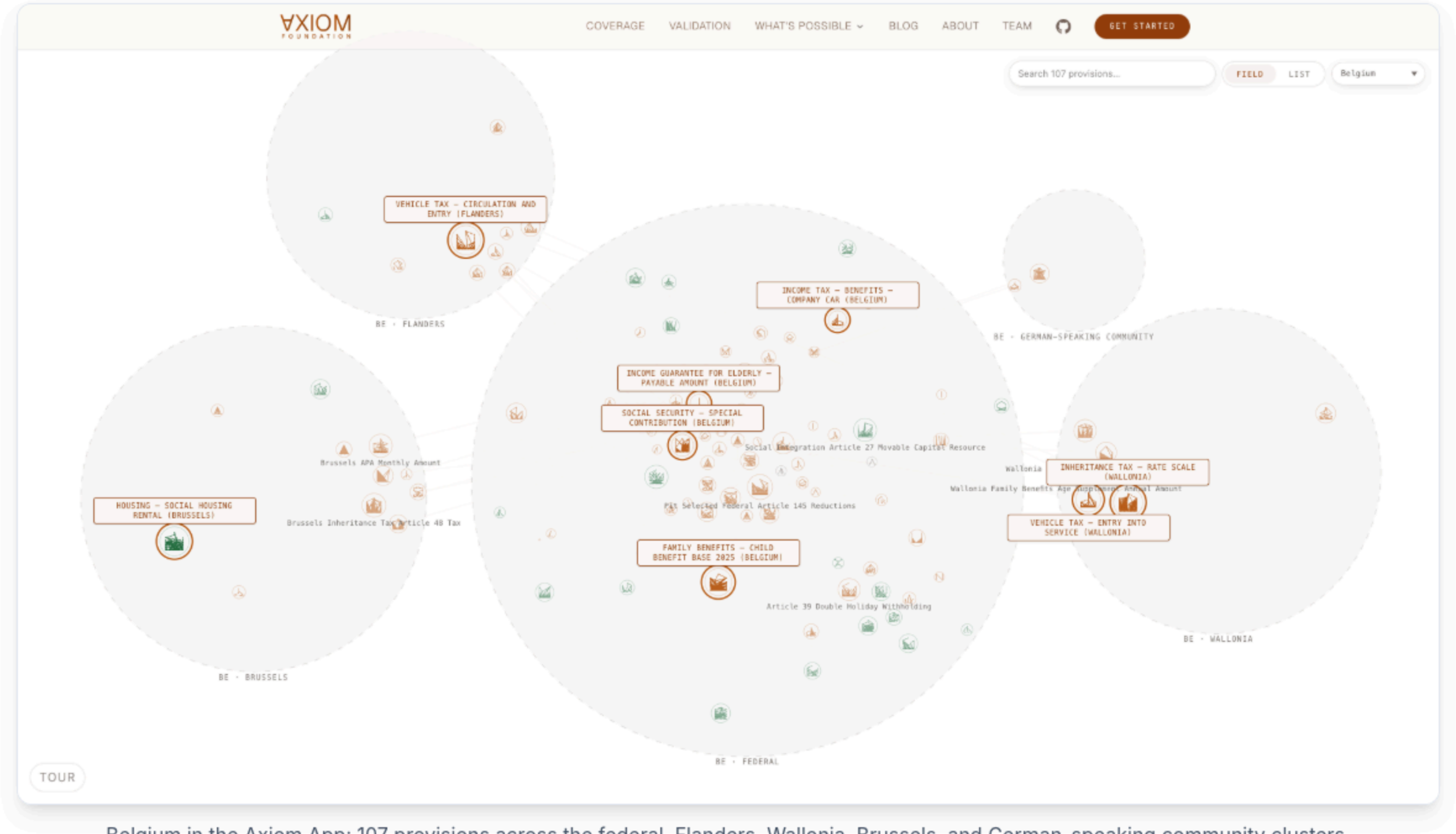
Every microplex release is reproducible end-to-end from public inputs. Calibration scores published openly, per release.

3 Live

Browse the law, run a reform, score a policy in
a few lines of Python.

Browse the encoded law

axiom.org



Belgium in the Axiom App: 107 provisions across the federal, Flanders, Wallonia, Brussels, and German-speaking community clusters.

The law at the node

AXIOM FOUNDATION

COVERAGE VALIDATION WHAT'S POSSIBLE BLOG ABOUT TEAM GET STARTED

THE LAW AT THIS NODE

AXIOM / BELGIUM / STATUTES / FISCONETPLUS / § CIR92 / (REVENUS-2025) / (PAGE-95)

§ 1. Les avantages de toute nature qui sont obtenus autrement qu'en espèces sont comptés pour la valeur réelle qu'ils ont dans le chef du bénéficiaire. Dans les cas qu'il détermine, le Roi peut fixer des règles d'évaluation forfaitaire de ces avantages. § 2. L'avantage de toute nature résultant de l'utilisation à des fins personnelles d'un véhicule mis gratuitement à disposition visé à l'article 65 est calculé en appliquant un pourcentage-CO2 à 6/7 de la valeur catalogue du véhicule mis gratuitement à disposition. Par valeur catalogue, il faut entendre le prix catalogue du véhicule à l'état neuf lors d'une vente à un particulier, options et taxe sur la valeur ajoutée réellement payée comprises, sans tenir compte des réductions, diminutions, rabais ou ristournes. La valeur catalogue est multipliée par un pourcentage qui est déterminé dans le tableau ci-dessous, afin de tenir compte de la période écoulée à partir de la date de la première immatriculation du véhicule: Période écoulée depuis la première immatriculation du véhicule (un mois commencé compte pour un mois entier) Pourcentage de la valeur catalogue à prendre en considération lors du calcul de l'avantage De 0 à 12 mois 100 % De 13 à 24 mois 94 % De 25 à 36 mois 88 % De 37 à 48 mois 82 % De 49 à 60 mois 76 % A partir de 61 mois 70 % Le pourcentage de base CO2 s'élève à 5,5 % pour une émission de référence-CO2 de 115 g/km pour les véhicules à moteur alimenté à l'essence, au LPG ou au gaz naturel et pour une émission de référence-CO2 de 95 g/km pour les véhicules à moteur alimenté au diesel. Le Roi détermine chaque année l'émission de référence-CO2 en fonction de l'émission CO2 moyenne sur une période de douze mois consécutifs se terminant le 30 septembre de l'année qui précède la période imposable par rapport à l'émission CO2 moyenne de l'année de référence 2011 selon les modalités qu'il fixe. L'émission CO2 moyenne est calculée sur la base de l'émission CO2 des véhicules visés à l'article 65 qui sont nouvellement immatriculés. L'émission de référence-CO2 ne peut jamais être supérieure à l'émission de référence-CO2 de la période imposable antérieure. Lorsque l'émission du véhicule concerné dépasse l'émission de référence précitée, le pourcentage de base est augmenté de 0,1 % par gramme de CO2, avec un maximum de 18 %. Lorsque l'émission du véhicule concerné est inférieure à l'émission de référence précitée, le pourcentage de base est réduit de 0,1 % par gramme de CO2, avec un minimum de 4 %. Les véhicules pour lesquels aucune donnée relative à l'émission de CO2 n'est disponible au sein de la direction de l'immatriculation des véhicules sont assimilés, s'ils sont propulsés par un moteur à essence, au

belgium_company_car_missing_petrol_lp_

be/statute/fisconetplus/cir92/revenus-2025/page-95 - para...

belgium_company_car_petrol_lpg_natura_

be/statute/fisconetplus/cir92/revenus-2025/page-95 - para...

belgium_company_car_raw_co2_percentage

be/statute/fisconetplus/cir92/revenus-2025/page-95 - deriv...

belgium_company_car_reference_co2

be/statute/fisconetplus/cir92/revenus-2025/page-95 - deriv...

belgium_company_car_reported_or_defau_

be/statute/fisconetplus/cir92/revenus-2025/page-95 - deriv...

belgium_company_car_taxable_

YOUR RULE

be/statute/fisconetplus/cir92/revenus-2025/page-95 - deriv...

CIR 1992 article 36 paragraph 2 beneficiary contribution rule

```
name: belgium_company_car_taxable_annual
kind: derived
entity: Person
dtype: Money
period: Year
unit: EUR
source: CIR 1992 article 36 paragraph 2
versions:
  - effective_from: '2026-01-01'
    formula: >=
```

OVERVIEW

RESULT

Belgium Company Car Age Adjusted Catalog Value

PARAMETER

Belgium Co Minimum An Assessment

PARAMETER

Belgium Company Car Catalog Value Fraction

TOUR

ALL COLLAPSE ALL Run a scenario

Belgium Company Car Taxable Annual Benefit

FROM

Belgium Company Car Taxable Annual Benefit

USED BY

nothing — a final result

DETAILS

FORMULA

te this rule

the law →

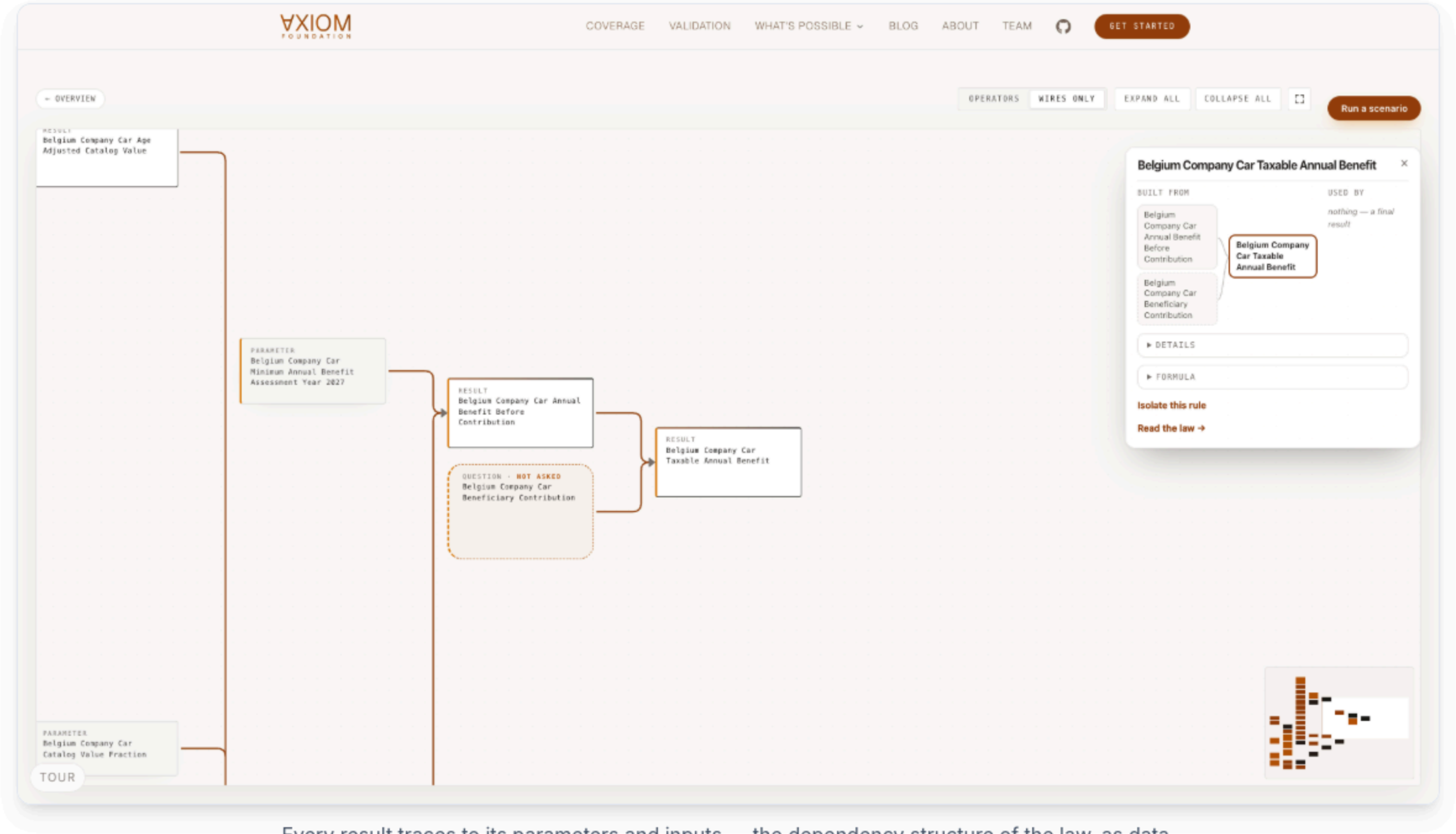
CIR 92, art. 36 §2 — the statute text beside the RuleSpec rule that computes it, with entity, unit, and effective dates.

POLICY ENGINE

PolicyEngine · IARIW-CAPE workshop · August 2026

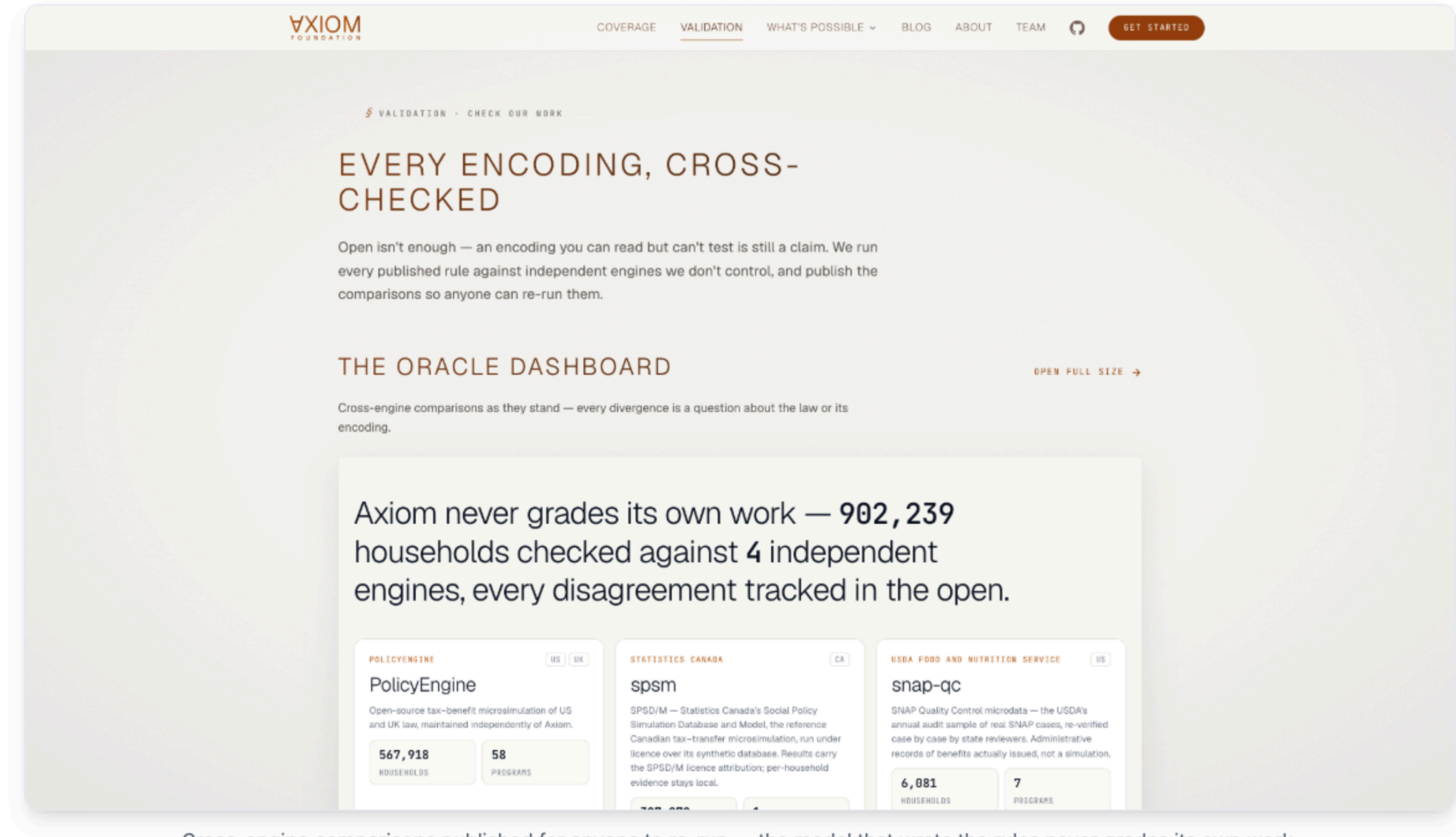
The computation graph

axiom.org



Every result traces to its parameters and inputs — the dependency structure of the law, as data.

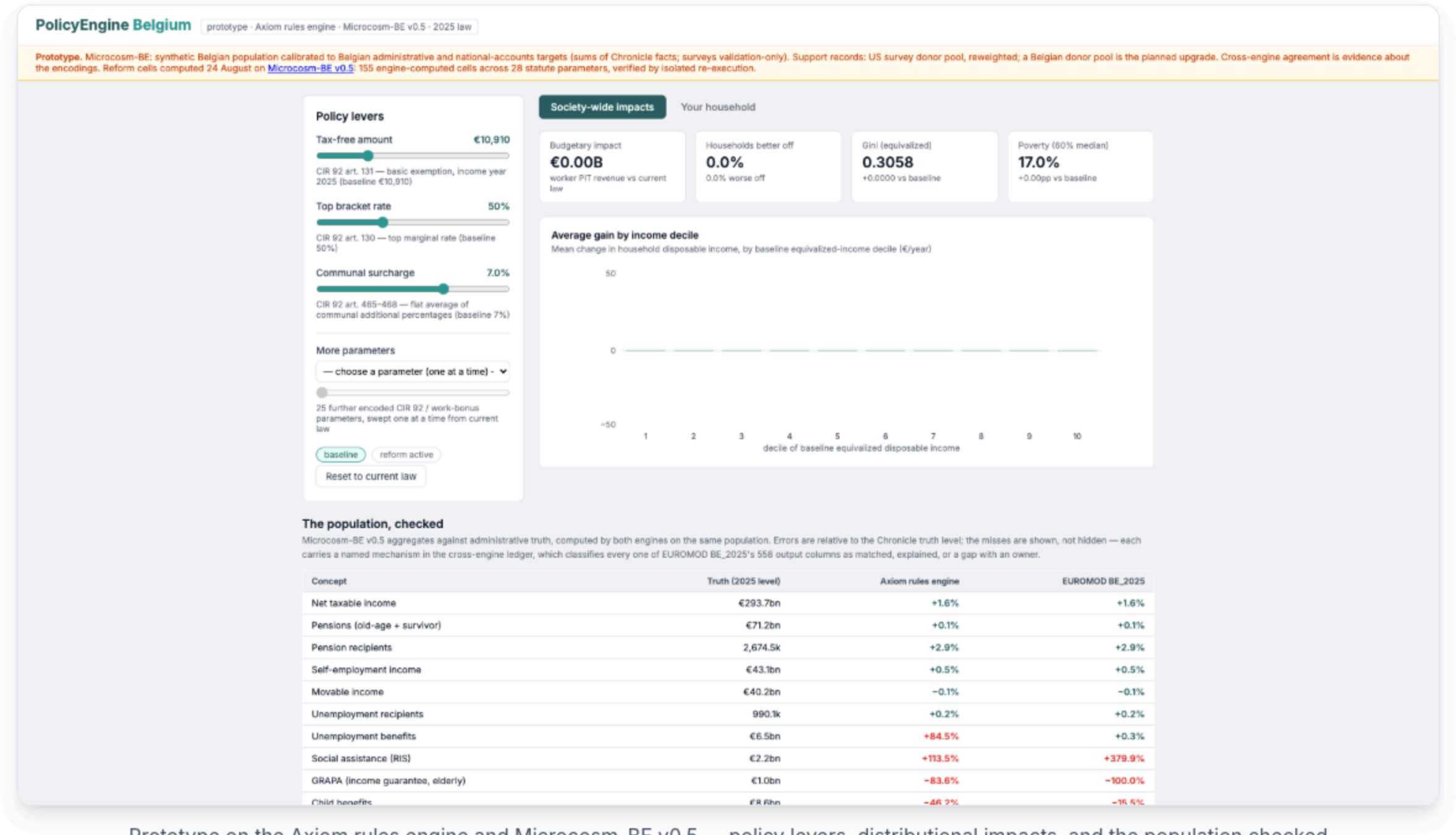
axiom.org/validation



Cross-engine comparisons published for anyone to re-run — the model that wrote the rules never grades its own work.

A Belgian reform, computed

policyengine.org/be



The same models, as a Python package

pip install policyengine — the interface behind the apps

```
policyengine.py

import policyengine as pe

# UK: single adult earning £50,000
uk = pe.uk.calculate_household(
    people=[{"age": 35, "employment_income": 50_000}],
    year=2026,
)
print(uk.person[0].income_tax)

# US: same call, with a reform attached
us = pe.us.calculate_household(
    people=[{"age": 35, "employment_income": 60_000}],
    tax_unit={"filing_status": "SINGLE"},
    household={"state_code": "CA"},
    year=2026,
    reform={"gov.irs.credits.ctc.amount.adult_dependent": 1000},
)
print(us.tax_unit.income_tax)
```

Households and populations

The same package runs a single household or a full calibrated population, baseline against reform.

Reforms are parameter changes

Every parameter carries its legislative reference; a reform is a dated override of the ones you name.

This is the interface new engines plug into — the Belgian work you just saw targets the same contract.

4

Where this goes

Conductors, uncertainty, and entering public administrations.

Conductors, not oracles

ORACLE (THE WRONG FRAME)

An LLM absorbs text and emits text. Confidently. Without calling the models that would actually answer the question.

This is the failure mode PolicyBench measures.

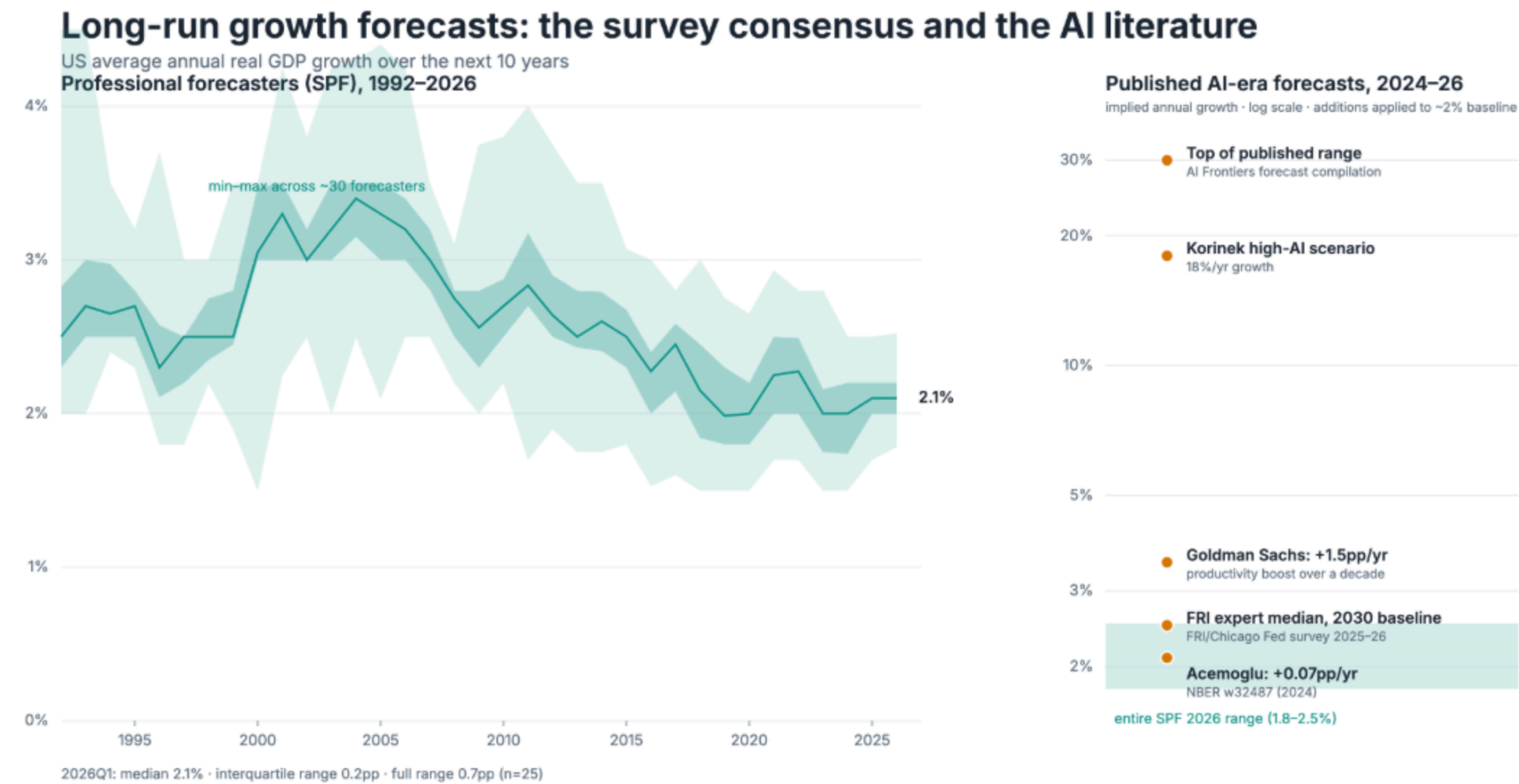
CONDUCTOR (THE RIGHT FRAME)

The model routes a question to the right tool — microsim, structural model, statistical method, archival lookup — with parameters, and integrates calibrated outputs.

Analyst judgment lives at every routing decision.

The judgment-to-mechanism loop: where LLM judgment captures something current models miss, that judgment eventually gets formalized into the next generation of models. Analyst intuition becomes mechanism over time — not the other way around.

Forecasters agree with each other — and the AI literature does not



The disagreement is not inside the survey — it lives outside it: a 0.7pp professional range against a 400× spread in the published AI literature (+0.07 to +30pp/yr).

Sources: Philadelphia Fed Survey of Professional Forecasters (RGDP10, individual responses, 2026Q1); Acemoglu, NBER w32487 (2024); Goldman Sachs Global Economics (2023); Forecasting Research Institute / Chicago Fed expert survey (Oct 2025–Feb 2026); AI Frontiers compilation of published forecasts; span runs Acemoglu +0.07pp/yr to Epoch AI +30pp/yr. Chart: PolicyEngine.

The professional 10-year consensus has never been tighter (IQR 0.2pp, half its 1990s level), while published AI-era growth estimates span +0.07pp to +30pp per year. Philadelphia Fed SPF; AI Frontiers compilation

When credible forecasts of the era span orders of magnitude, evidence infrastructure's job shifts from defending one number to letting a ministry recompute the present.

It takes all of us



Model-building has always been about making the consequences of policy visible before the choice is made.

AI lets this community produce more evidence — at more scale, more conditionally, more calibrated against what governments publish.

The work is not choosing the policy.

The work is knowing what each policy will actually do — so the people who do choose can align outcomes with what they are trying to achieve.

Statistical offices, EUROMOD, BEAMM, ministries, open models — each piece, doing what it does best.

It is the work of a community, not any one institution.

Thank you

See you at the roundtable at 16:00



axiom.org — browse the encoded law

axiom.org

policyengine.org/be

github.com/PolicyEngine · github.com/TheAxiomFoundation